

INSTITUTIONAL QUANTITATIVE RESEARCH & ARCHITECTURE

BLACKBOX X — PHASE 3.9

REGIME-AWARE PROBABILISTIC INTELLIGENCE METHODOLOGY
— Mathematical Formulation, Algorithmic Architecture, and Non-Predictive Research Guidelines — 1.1 Objective — 1.2 Non-Predictive Institutional Boundary

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BLACKBOX X Engine

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REGIME-AWARE PROBABILISTIC INTELLIGENCE METHODOLOGY

Mathematical Formulation, Algorithmic Architecture, and Non-Predictive Research Guidelines

1. Research Charter & Core Quantitative Invariant

1.1 Objective

Phase 3.9 extends BLACKBOX X from unconditional Monte Carlo simulations into **regime-conditioned probabilistic research**. The core research inquiry is:

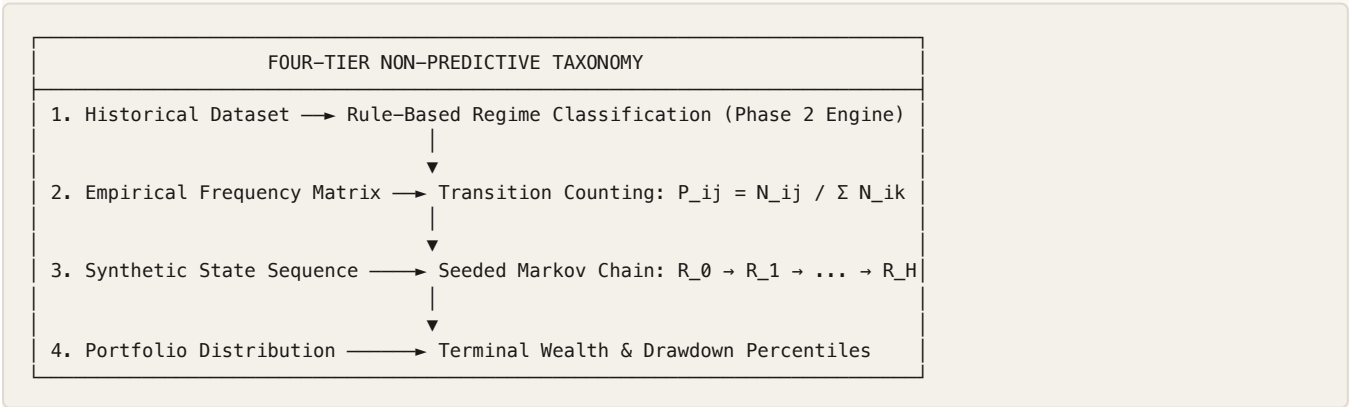
"How does the simulated distribution of portfolio outcomes change when the simulation is conditioned on observed market regimes and empirically observed regime transitions?"

1.2 Non-Predictive Institutional Boundary

BLACKBOX X **does not forecast future market regimes**, claim predictive foresight, or calculate "probabilities of crashes."

The system strictly delineates four analytical layers:

- Historical Observation:** Post-hoc rule-based regime classifications applied to historical daily price bars in the offline demonstration dataset.
- Empirical Transition Estimation:** The observed transition counts N_{ij} and frequencies P_{ij} between consecutive daily historical regime states.
- Simulated Regime Path:** A synthetic discrete-time Markov sequence $R_0 \rightarrow R_1 \rightarrow \dots \rightarrow R_H$ governed strictly by empirical transition frequencies.
- Simulated Portfolio Outcome:** The resulting terminal wealth trajectory and drawdown distribution conditioned on the sampled regime sequence.



Forbidden Phrasing:

- "the market will enter..."
- "the next regime will be..."
- "probability of the next crash..."

- "forecasted regime"

Approved Framing:

- "BLACKBOX X evaluates how portfolio outcome distributions differ under regime-conditioned simulation assumptions."
- "Empirical transition frequency"
- "Fraction of simulated paths"
- "Starting regime scenario"

2. Empirical Transition Matrix Formulation

Let $t \in \{1, \dots, T\}$ index the 1,825 synchronized contemporaneous daily observations ($T = 1,825$) spanning 2019-01-01 to 2023-12-31.

The platform defines four macro regimes:

$$\mathbf{R} = \{\text{BULL}, \text{BEAR}, \text{HIGH_VOLATILITY}, \text{LOW_VOLATILITY}\}$$

2.1 Transition Accumulation

For consecutive daily observation pairs (R_t, R_{t+1}) where $t \in \{1, \dots, T-1\}$:

1. **Transition Count Matrix $\mathbf{N} \in \mathbb{N}^{4 \times 4}$:**

$$N_{ij} = \sum_{t=1}^{T-1} \mathbf{I}(R_t = i \wedge R_{t+1} = j) \quad \forall i, j \in \mathbf{R}$$

2. **Total Outgoing Transitions:**

$$N_i = \sum_{j \in \mathbf{R}} N_{ij}$$

3. **Empirical Transition Probabilities $\mathbf{P} \in [0, 1]^{4 \times 4}$:**

$$P_{ij} = \begin{cases} \frac{N_{ij}}{N_i}, & N_i > 0 \\ 0, & N_i = 0 \end{cases}$$

2.2 Row-Stochastic Invariant

For every originating regime i with $N_i > 0$:

$$\sum_{j \in \mathbf{R}} P_{ij} = 1.0$$

No Artificial Smoothing: Arbitrary Laplace smoothing (+1 pseudocounts) or synthetic transitions are strictly prohibited. Transition probabilities represent empirical observation counts from the finite historical demonstration window.

3. Sparse Regime & Zero Outgoing Transition Policy

To maintain statistical integrity and numerical safety:

1. Minimum Observation Threshold:

$$\text{MIN_REGIME_OBSERVATIONS} = 20$$

If $T_i < 20$, the regime is flagged as `SPARSE\_WARNING`.

2. Zero Outgoing Transition Blocking: If an originating regime i has $N_i = 0$:

- The regime state is flagged as `'INSUFFICIENT_TRANSITION_DATA'`.
- The simulation engine **strictly blocks** simulations attempting to start in regime i , returning an explicit error: `"Cannot execute regime-conditioned Monte Carlo: Starting regime [i] has zero observed outgoing transitions (INSUFFICIENT_TRANSITION_DATA)."`
- During multi-regime comparative tables, regimes with $N_i = 0$ are rendered as `N/A (Insufficient Data)` rather than crashing the overall comparison.

4. Conditional Return Pools & Vector Preservation

For each regime $r \in \mathbf{R}$, observations are partitioned into conditional subsets:

$$D_r = \{\mathbf{r}_t \mid t \in \{1, \dots, T\} \wedge R_t = r\}$$

4.1 Indivisible Multi-Asset Vector Invariant

For every day $t \in D_r$, the return vector $\{\text{BTC}, t\}, r_{\{\text{NVDA}\}, t}^{\text{Trt}} = [\text{rGOLD}, t, \text{rBTC}, t, \text{rNVDA}, t]^T$ is treated as an indivisible atomic tuple. Assets are never independently bootstrapped within a regime, preserving authentic cross-asset dependence and flight-to-safety dynamics.

4.2 Conditional Moments

1. Conditional Mean Vector $\boldsymbol{\mu}_r$:

$$\boldsymbol{\mu}_r = \frac{1}{T_r} \sum_{t \in D_r} \mathbf{r}_t$$

2. Conditional Daily Covariance $\boldsymbol{\Sigma}_r$:

$$\boldsymbol{\Sigma}_r = \frac{1}{T_r - 1} \sum_{t \in D_r} (\mathbf{r}_t - \boldsymbol{\mu}_r)(\mathbf{r}_t - \boldsymbol{\mu}_r)^T$$

3. Annualized Covariance & Volatility:

$$\boldsymbol{\Sigma}_{r,\text{annual}} = 252 \times \boldsymbol{\Sigma}_r, \quad \sigma_{r,i} = \sqrt{\Sigma_{r,ii}} \times \sqrt{252} \times 100\%$$

5. Simulation Methods

5.1 Method A: Regime-Conditioned Bootstrap (`REGIME_BOOTSTRAP`)

At forward day $\tau \in \{1, \dots, H\}$:

1. Identify active regime R_τ .
2. Uniformly sample index $k \sim \mathcal{U}\{1, T_{R_\tau}\}$.
3. Resample synchronized vector:

$$\mathbf{r}_\tau^* = \mathbf{r}_k \in \mathcal{D}_{R_\tau}$$

4. Update portfolio positions and evaluate rebalance triggers.
5. Transition regime state at close of day τ for day $\tau + 1$:

$$R_{\tau+1} \sim \mathbf{P}_{R_\tau}.$$

5.2 Method B: Regime-Conditioned Parametric (`REGIME_PARAMETRIC`)

At forward day $\tau \in \{1, \dots, H\}$:

1. Retrieve conditional moments $\boldsymbol{\mu}_{R_\tau}$ and lower-triangular Cholesky factor $\mathbf{L}_{R_\tau}$.
 - If decomposition fails positive-definiteness ($L_{jj}^2 \leq 10^{-10}$), reactive Tikhonov stabilization is applied: $\boldsymbol{\Sigma}_{r,\text{stable}} = \boldsymbol{\Sigma}_r + \lambda_r \mathbf{I}$.
2. Generate standard normal shocks: $\mathbf{z}_\tau \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ via Box-Muller transform.
3. Compute correlated shocks: $\boldsymbol{\varepsilon}_\tau = \mathbf{L}_{R_\tau} \mathbf{z}_\tau$.
4. Simulated return vector: $\mathbf{r}_\tau^* = \boldsymbol{\mu}_{R_\tau} + \boldsymbol{\varepsilon}_\tau$.
5. Update portfolio positions with bankruptcy clamping ($r_i^* \leq -100\% \rightarrow V_{i,\tau} = 0$).
6. Transition regime state for day $\tau + 1$.

6. Starting Regime Semantics & Day-1 Protocol

The simulation supports six starting regime modes:

1. `START_CURRENT_OBSERVED` : Starts in the latest historical state (R_T , observation 1,825).
2. `START_BULL` : Initial regime $R_0 = \text{BULL}$.
3. `START_BEAR` : Initial regime $R_0 = \text{BEAR}$.
4. `START_HIGH_VOL` : Initial regime $R_0 = \text{HIGH_VOL}$.
5. `START_LOW_VOL` : Initial regime $R_0 = \text{LOW_VOL}$.
6. `START_EMPIRICAL_DISTRIBUTION` : Initial regime R_0 sampled from observed occupancy frequencies $\boldsymbol{\pi} = [T_{\text{BULL}}/T, \dots, T_{\text{LOW_VOL}}/T]$.

Day-1 Return Convention Invariant:

- Return at forward step $\tau = 1$ is generated **directly from the starting regime** R_0 .
- The transition $R_0 \rightarrow R_1$ occurs at the close of day 1, governing returns for day 2.

7. Regime Persistence & Dwell Time Diagnostics

1. Occupancy Frequency:

$$\pi_i = \frac{T_i}{T} \in [0, 1], \quad \sum_{i=1}^4 \pi_i = 1.0$$

2. Empirical Mean & Median Dwell Time:

- $\bar{d}_i = \frac{1}{K_i} \sum_{k=1}^{K_i} d_{i,k}$ consecutive days.
- $\text{Median}(d_i)$.

3. Self-Transition Stickiness:

$$P_{ii} = \frac{N_{ii}}{N_i}$$

4. Theoretical Geometric Dwell Time:

$$E[D_i] = \frac{1}{1 - P_{ii}} \quad (\text{for } P_{ii} < 1.0)$$

5. Switching Frequency:

$$\text{Rate}_{\text{switch}} = \frac{\sum_{i \neq j} N_{ij}}{T - 1}$$

8. Analytical Separation: Three-Tier Hierarchy

THREE-TIER QUANTITATIVE HIERARCHY		
Tier	Simulation Mechanism	Research Question Answered
A. Historical Backtest	Realized 2019–2023 Historical Daily Bars	What actually occurred over the common synchronized bar?
B. Unconditional Monte Carlo (3.8)	Stationary Resampling Without Regime Shifts	What is outcome dispersion under stationary history?
C. Regime-Conditioned Monte Carlo (3.9)	Markov Regime Shifts + Conditional Pools	How do outcome distributions shift by macro regime?

9. Performance & Bounded Execution Architecture

- Web Worker Non-Blocking Execution:** Reuses `monteCarlo.worker.ts` with dedicated message protocol (`action: 'REGIME'`).
- Memory Ceiling:**
 - Streaming percentile reductions (P05, P25, P50, P75, P95).
 - Up to 101 trajectory fan coordinates.
 - Exactly 50 representative sample paths transferred to React state.

- Net message payload < 5 MB.

3. **Deterministic PRNG:** 32-bit Mulberry32 generator ensures bit-identical reproducibility for identical configurations and seeds across both Web Worker and direct Node.js executions.